Is Photosynthesis Nutrient Limited in Tropical Trees?

Louis S. Santiago and Guillermo Goldstein

Abstract Tropical forests play an enormously important role in the global cycling of carbon. However, the extent to which nutrients limit the potential for tropical trees to increase carbon gain as atmospheric carbon dioxide rises, a phenomenon known as the carbon-concentration feedback, is uncertain. This chapter addresses our current state of knowledge on nutrient limitation of photosynthesis in tropical trees, summarizing and synthesizing the results of over 20 studies on photosynthetic responses to nutrient manipulation experiments. Our results indicate that nutrient limitation of photosynthesis is widespread, but that contrasting species and ecosystems vary in their responses, with savannah trees showing the least response at the leaf scale. Second, although photosynthesis is strongly limited by N in particular species, N limitation of photosynthesis is modest compared to P limitation of photosynthesis when considering all of the available literature. Finally, alleviation of nutrient limitation through addition of combined nutrient treatments produces the strongest photosynthetic responses, highlighting the potentially complex stroichiometric interactions among elements. The authors discuss several ways forward for resolving questions regarding the potential and limits of tropical trees to influence key carbon cycling processes and thus improve our ability to forecast global responses and feedbacks to climate change.

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Introduction

Tropical forests are responsible for approximately one-third of terrestrial gross primary production (Beer et al. 2010), and are thought to be especially sensitive to climate change (Phillips et al. 2010), suggesting that their responses to climate change and feedbacks to the global climate system are likely to be substantial. But one outstanding question is whether tropical trees can exploit the increasing atmospheric supply of CO_2 through enhanced photosynthetic carbon income (the concentration-carbon feedback) (Bonan and Levis 2010). Understanding the trajectory of plant photosynthetic responses to increasing CO_2 is not straightforward because nutrient availability has the potential to limit the concentration-carbon feedback in tropical vegetation such that photosynthesis may not continue to increase with atmospheric CO_2 concentration (Santiago 2015). Predictions are further complicated by the fact that soil substrates vary greatly in the tropics. For example, the Amazon lies along a dramatic east–west nutrient availability gradient (Quesada et al. 2010), and there are often strong changes in nutrient availability with elevation (Dalling et al., this volume). So how does the formidable diversity of trees combine with heterogeneity of soil types and elemental availability in the tropics to limit photosynthetic exploitation of increasing atmospheric CO_2? To address this question, this synthesis focuses on photosynthetic responses of tropical trees to nutrients and how such information is critical to inform Earth system models to understand potential climate change feedbacks.

There are multiple ways in which nutrients could mediate the concentration-carbon feedback. The first is simply that plants increase leaf-level photosynthetic rates per area with increasing atmospheric CO_2 to the degree allowed by nutrient availability (Bloom et al. 1985; Field and Mooney 1986) (Fig. 1a). The second is that as atmospheric CO_2 increases, plants allocate increased carbon income to growth such that leaf-level photosynthesis per area remains the same, but the greater leaf surface area associated with enhanced growth leads to increased photosynthesis at the whole-plant scale (Poorter and Remkes 1990) (Fig. 1b). A third scenario is one in which increasing CO_2 causes a decrease in stomatal conductance such that internal CO_2 concentration and photosynthetic rate remain the same, but water use efficiency, defined as photosynthetic carbon gain per unit water loss, increases (Fig. 1c) (Farquhar and Richards 1984; Cernusak et al. 2007a, b). In all of these scenarios we expect nutrients to mediate physiological responses to changing atmospheric CO_2 (Raven et al. 2004; Cernusak et al. 2007a, b), such that tropical trees growing on the most fertile soils would have the greatest increase, or least reduction in photosynthetic carbon assimilation. All of these potential responses to atmospheric CO_2 could also have larger scale impacts because of the strong bearing

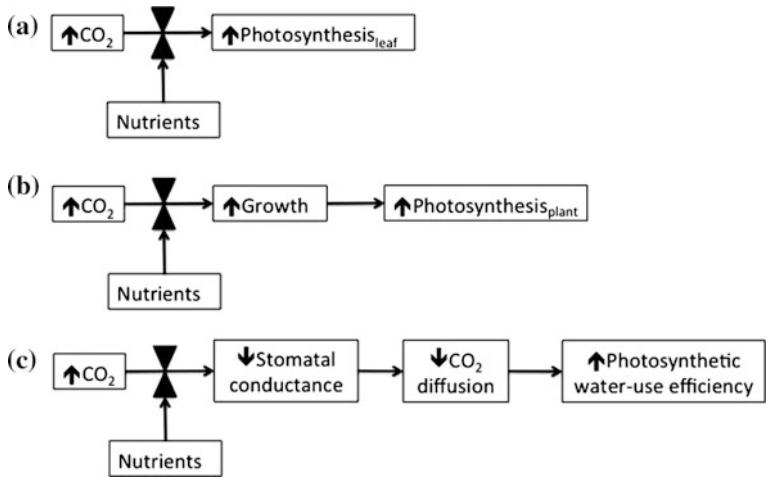


Fig. 1 Possible ways for nutrients to mediate responses of tropical plants to increasing CO_2

of tropical forests on the global carbon cycle and climate system (Friedlingstein et al. 2006; Bonan and Levis 2010). Yet the only way to determine the extent to which nutrients limit the carbon-concentration feedback in tropical trees is to determine the identity of elements that limit photosynthesis and the degree of that limitation.

There are also several valid eco-physiological reasons why tropical woody species may not show greater photosynthesis on more fertile soils. First, several studies have now reported increased herbivory following nutrient addition and an increase in foliar nutrient concentration that appears to make leaves more palatable to herbivores (Campo and Dirzo 2003; Santiago et al. 2012). In some cases, the increase in herbivory can be strong enough to mask plant growth responses to nutrient addition (Andersen et al. 2010). Second, there is evidence that as atmospheric CO_2 increases, some plants reduce stomatal conductance such that photosynthetic rates remain the same, but water loss goes down and water use efficiency goes up (Cernusak et al. 2013). Because nutrient availability can interact with plant regulation of water loss, especially when transpiration rates are linked to uptake of key elements through mass flow in the transpiration stream (Cramer et al. 2009), reductions in stomatal conductance with increasing CO_2 could reduce nutrient uptake through mass flow and worsen nutrient limitation of photosynthesis. In other cases, plants in poor soils have been shown to exhibit greater transpiration rates than plants in fertile soils and it is thought that the benefit is increased rates of absorption of nutrients through mass flow (Cramer et al. 2008). In such cases, how plants in poor versus fertile soils might alter their photosynthetic rates with changing atmospheric CO_2 is unknown. Finally, if increasing atmospheric CO_2 concentration indeed increases photosynthetic carbon uptake, at some point a buildup of carbohydrate products of photosynthesis could lead to down regulation

of photosynthetic rate in a process known as feedback inhibition (Delucia et al. 1985). Thus careful evaluation of the study design, including plant-animal interactions and other physiological processes such as nutrient allocation to leaves and interactions with transpiration will increase the depth of interpretation and predictive power of experimental studies.

How Is Nutrient Limitation Studied?

Nutrient limitation of plant productivity and photosynthesis has been studied through a variety of approaches. Perhaps the most effective has been through nutrient addition experiments in which responses to fertilization relative to control treatments are interpreted as limitation by that nutrient, of if there is no response, the conclusion is that the added nutrient was not limiting (Vitousek 2004). Many of these are stand-level fertilization experiments consisting of large (40 × 40 m) plots receiving individual or combined addition of nitrogen (N), phosphorus (P), potassium (K) or micronutrients (Tanner et al. 1992; Vitousek et al. 1993; Mirmanto et al. 1999; Newbery et al. 2002; Campo and Vázquez-Yanes 2004; Davidson et al. 2004; Bucci et al. 2006; Lu et al. 2010; Wright et al. 2011; Álvarez-Clare et al. 2013; Fisher et al. 2013). Such large experiments are able to evaluate nutrient limitation at multiple scales including leaf physiological processes, growth and ecosystem-level processes such as nutrient cycling and litterfall. Yet, because canopy access poses substantial challenges and because ecosystem science is usually the focus of these studies, measurements of leaf-scale physiological measurements are only available from five of the known large-scale tropical fertilization experiments (Cordell et al. 2001; Saraceno 2006; Lu et al. 2007; Pasquini and Santiago 2012).

Other approaches to manipulating nutrient availability in tropical trees include litter removal and addition experiments (Sayer and Tanner 2010; Sayer, this volume) or removal of vegetation to increase nutrient availability of target individuals. Such process-based nutrient manipulations do not directly reveal the identities of limiting elements. However, when foliar nutrient responses are included, information on the relative importance of key elements can sometimes be determined (Lewis and Tanner 2000; Barberis and Tanner 2005). At smaller scales, fertilizer has also been applied directly to individual plants in the field (e.g. Yavitt and Wright 2008), or in pots (Burslem 1996; Gunatilleke et al. 1997; Lawrence 2003; Cai et al. 2008). In many cases, such experiments have been successful in determining which elements limit growth and physiological processes in tropical seedlings or saplings.

When stand-scale experimental manipulation of nutrient availability is impractical, basic questions about plant nutritional effects on physiology have been approached along environmental gradients that vary in soil nutrient availability (Vitousek et al. 1992; Santiago et al. 2004; Santiago and Mulkey 2005; Baltzer et al. 2008; Fyllas et al. 2009). In these in situ studies, by evaluating how foliar

nutrient concentration or physiological characteristics vary with soil type it is possible to generate hypotheses regarding nutrient limitation that can be further tested experimentally. For example, in Hawaii, the gradual shift from N-limitation of ecosystem productivity in very young soils to P-limitation of ecosystem productivity in very old soils creates opportunities to evaluate the relative importance of these elements for photosynthesis and growth (Kitayama et al. 1997; Cordell et al. 2001). In lowland tropical forest in Panama, increasing precipitation leads to species replacement of dry-season deciduous tree species with evergreen species, consistent with the idea that evergreeness conserves nutrients by minimizing the cost of leaf replacement and providing an advantage for evergreen species in wetter sites with leached soils (Santiago and Mulkey 2005). Even broad ecological patterns, such as in Brazilian savannahs, where patches of closed canopy dry season deciduous forest appear on calcium-rich outcrops (Haridasan 1992), provide fodder for new hypotheses and experiments. Thus the interplay between observation and experiment has continued to be an important process in understanding nutrient limitation in the tropics.

Direct analysis of foliar ratios of nitrogen-to-phosphorus (N:P) is also considered a method for inferring relative nutrient limitation of N versus P using the guideline that N:P ratios below 12 indicate N-limitation, ratios above 16 indicate P-limitation and values between 12 and 16 indicate co-limitation by N and P (Aerts and Chapin 2000). It is difficult for foliar N:P data alone to establish limitation, yet it seems to be an excellent measure of relative limitation. Previously, most researchers used the eco-physiological standard of the youngest mature leaves to study N:P, but recently, stem, root, and older leaf N:P have been shown to be more responsive indicators of soil nutrient availability than new leaf N:P, because new leaves appear to act as nutrient reservoirs during active growth, allowing maintenance of optimal N:P ratios in recently produced, physiologically active leaves (Schreeg et al. 2014). Overall, combining soil and plant observational approaches with experiments offer our greatest potential for unraveling complex interactions between nutrient availability, increasing atmospheric CO_2 and photosynthetic performance in tropical forests.

The Physiological Basis for Elemental Limitation of Photosynthesis

The proteins of the Calvin cycle and thylakoids make up the majority of N in leaves, thus N is known to be a strong determinant of photosynthesis (Field and Mooney 1986; Wright et al. 2004). Therefore, as leaf N concentration increases with experimental fertilization (Ostertag 2009; Wright et al. 2011; Pasquini and Santiago 2012; Santiago et al. 2012), maximum photosynthetic rate is expected to increase as well. N addition in shade leaves is expected to increase N allocation toward light harvesting proteins of the thylakoid membrane of the chloroplast,

including the pigment-protein complexes and components of the electron transport chain (Evans 1989). In sun leaves, the maximum rate of CO_2 assimilation (A_{max}) is expected to increase with N addition, but in any case, there should be correspondence between light harvesting and carboxylation because carboxylation capacity and electron transport are tightly coupled (Wullschleger 1993; Pasquini and Santiago 2012). The increase in foliar N with fertilization is usually modest, owing to the metabolic costs of storing N in plant tissue (Ostertag 2009). Thus the expected enhancement of photosynthesis with N addition should occur over a relatively small range of leaf N values.

Leaf P is a component of polyphosphates and phospholipids and is necessary for activity of the Calvin cycle and rubisco regeneration (Marschner 1995). Within the biochemical reactions of carbon assimilation, activation through phosphorylation is ubiquitous, highlighting the potential for P to limit or promote rapid photosynthetic reactions. P addition has been shown to stimulate A_{max}, but reduce A_{max} per unit leaf P (Cordell et al. 2001), because of the ability of leaves to sequester large amounts of P in vacuoles (Sinclair and Vadez 2002; Ostertag 2009). In most soils of lowland tropical forests, P is thought to be the most limiting element to productivity (Dalling, this volume), so understanding P limitation of photosynthesis is fundamental for predicting plant responses to climate change.

K^+ is the most abundant ion in plant cells and is critical for numerous biochemical functions including osmoregulation, photosynthesis, cell extension, oxidative phosphorylation and protein activation (Evans and Sorger 1966; Morgan 1984; Marschner 1995; Santiago and Wright 2007). Therefore, K addition has the potential to increase photosynthesis through efficient biochemistry and stomatal control, but of the macronutrients, it is the least studied and most rare to find in fertilization experiments on tropical trees. Currently, there is only one stand-scale natural forest fertilization study that has incorporated K, and growth of seedlings, saplings and poles, but not emergent trees, as well as aspects of photosynthesis have been shown to be limited by K (Wright et al. 2011; Pasquini and Santiago 2012). K is also unique in that it has no organic phase, so unlike N and P which can be bound to organic complexes within leaf litter, K leaches quickly (Schreeg et al. 2013), potentially promoting shortages in availability.

Nutrient Addition Studies with Tropical Trees

Of the nutrient addition studies on tropical plants, we have identified 19 studies on 46 species that measured A_{max} on plants fertilized with N, P, K or a combination of these and micronutrient elements (Table 1). These include studies on plants in pots in controlled greenhouse settings, as well as plants in the field, growing in large stand-level fertilized or control plots. We only included studies that report A_{max} for individual species at multiple nutrient levels, but did not consider proxy variables

Table 1 Summary of species, locations, experimental details and references for studies involving responses of maximum photosynthetic rate to nutrient addition in tropical tree species

Location	Species	Treatment	Study type	References
Brasília, Brazil	*Blepharocalyx salicifolius*	N, P, Combined	field	Saraceno (2006)
Brasília, Brazil	*Caryocar brasiliense*	N, P, Combined	field	Saraceno (2006)
Brasília, Brazil	*Ouratea hexasperma*	N, P, Combined	field	Saraceno (2006)
Brasília, Brazil	*Qualea parviflora*	N, P, Combined	field	Saraceno (2006)
Dinghushan, China	*Cryptocarya chinensis*	N	field	Lu et al. (2007)
Dinghushan, China	*Cryptocarya concinna*	N	field	Lu et al. (2007)
Dinghushan, China	*Randia canthioides*	N	field	Lu et al. (2007)
Dinghushan, China	*Cryptocarya concinna*	N, P, Combined	field	Zhu et al. (2014)
Dinghushan, China	*Randia canthioides*	N, P, Combined	field	Zhu et al. (2014)
Freetown, Sierra Leone	*Entandrophragma angolense*	Combined	pot	Riddoch et al. (1991)
Freetown, Sierra Leone	*Nauclea diderrichii*	Combined	pot	Riddoch et al. (1991)
Fujian, China	*Eucalyptus dunnii*	P	pot	Wu et al. (2014)
Fujian, China	*Eucalyptus grandis*	P	pot	Wu et al. (2014)
Gamboa, Panama	*Ficus insipida*	Combined	pot	Cernusak et al. (2007a, b)
Gamboa, Panama	*Platymiscium pinnatum*	Combined	pot	Cernusak et al. (2009)
Gamboa, Panama	*Swietenia macrophylla*	Combined	pot	Cernusak et al. (2009)
Gamboa, Panama	*Tectona grandis*	Combined	pot	Cernusak et al. (2009)
Gigante, Panama	*Alesis blackiana*	N, P, K	field	Pasquini and Santiago (2012)
Guangzhou, China	*Acmena acuminatissima*	N	chamber	Liu et al. (2012)
Guangzhou, China	*Castanopsis hystrix*	N	chamber	Liu et al. (2012)
Guangzhou, China	*Ormosia pinnata*	N	chamber	Liu et al. (2012)
Guangzhou, China	*Schima superba*	N	chamber	Liu et al. (2012)
Hawaii, USA	*Metrosideros polymorpha*	N, P, Combined	field	Cordell et al. (2001)
Queensland, Australia	*Auranticarpa ilicifolia*	P	pot	Bloomfield et al. (2014)
Queensland, Australia	*Cryptocarya triplinervis*	P	pot	Bloomfield et al. (2014)
Queensland, Australia	*Flindersia bourjotiana*	P	pot	Bloomfield et al. (2014)
Queensland, Australia	*Guioa lasioneura*	P	pot	Bloomfield et al. (2014)
Queensland, Australia	*Hymenosporum flavum*	P	pot	Bloomfield et al. (2014)

(continued)

Table 1 (continued)

Location	Species	Treatment	Study type	References
Queensland, Australia	*Neolitsea dealbata*	P	pot	Bloomfield et al. (2014)
Queensland, Australia	*Syzygium wilsonii*	P	pot	Bloomfield et al. (2014)
Queensland, Australia	*Flindersia brayleyana*	Combined	pot	Thompson et al. (1988)
Queensland, Australia	*Argyrodendron trifoliolatum*	Combined	pot	Thompson et al. (1992)
Queensland, Australia	*Argyrodendron sp*	Combined	pot	Thompson et al. (1992)
Queensland, Australia	*Flindersia brayleyana*	Combined	pot	Thompson et al. (1992)
Queensland, Australia	*Toona australis*	Combined	pot	Thompson et al. (1992)
Recife, Brazil	*Anadenanthera colubrina*	P	pot	Oliveira et al. (2014)
Recife, Brazil	*Prosopis juliflora*	P	pot	Oliveira et al. (2014)
Sabah, Malaysia	*Dryobalanops lanceolata*	Combined	field	Bungard et al. (2002)
Sabah, Malaysia	*Shorea johorensis*	Combined	field	Bungard et al. (2002)
San Antonio, Costa Rica	*Cedrela odorata*	Combined	pot	Carswell et al. (2000)
Sao Paulo, Brazil	*Cariniana legalis*	Combined	pot	de Oliveira et al. (2012)
Sao Paulo, Brazil	*Croton urucurana*	Combined	pot	de Oliveira et al. (2012)
Twin Cays, Belize	*Rhizophora mangle*	N, P	field	Lovelock et al. (2006)
Xishuangbanna, China	*Bauhinia aurea*	N	pot	Cai et al. (2008)
Xishuangbanna, China	*Bauhinia claviflora*	N	pot	Cai et al. (2008)
Xishuangbanna, China	*Bauhinia monandra*	N	pot	Cai et al. (2008)
Xishuangbanna, China	*Bauhinia purpurea*	N	pot	Cai et al. (2008)
Xishuangbanna, China	*Bauhinia tenuiflora*	N	pot	Cai et al. (2008)

Reference (Oliveira et al. 2014)

related to photosynthetic rate, such as foliar nutrient concentration or specific leaf area. Whereas most of these studies used simple fertilizer additions, four created Hoagland or Ingestad solutions to match desired levels of nutrient availability (Thompson et al. 1988, 1992; Riddoch et al. 1991; Carswell et al. 2000), but in an attempt to be inclusive we considered all of these studies due to the limited availability of such information in the literature. Within Table 1, there is also

variation in light availability among studies so we used the highest light availability if there were multiple levels, yet some studies (Lu et al. 2007; Pasquini and Santiago 2012), were done on understory plants in the field, so very low light conditions prevailed. Similarly, in studies where multiple levels of nutrient addition were performed, we utilized treatments with the highest level of nutrient addition except for the study of Lu et al. (2007), in which the goal of the study was to simulate N deposition to the point of toxicity. Here, we used the lowest level of fertilization, which was similar to the high nutrient treatments in other studies. If multiple time points were given in the paper, we used the latest time point to represent the values produced under a longer period of experimental treatment. Finally, we restricted our analysis to woody species and included lianas in addition to trees, thus three of the *Bauhinia* species in Table 1 are lianas (*B. claviflora, B. tenuiflora, B. aurea*).

Our data show that overall there have been enhancements of A_{max} with nutrient addition in tropical trees (Fig. 2). Of the studies on nutrient limitation of A_{max} in this synthesis, 22 included control and high N treatments, 21 included control and high P treatments, and 24 included control and combined elemental treatments of various kinds, for a total of 67 species × site combinations (Table 1). The raw data shows that mean A_{max} trended higher with N, P or combined nutrient addition (Fig. 2). However, to evaluate the overall effect of nutrient addition on photosynthesis, we used the *ln*-transformed response ratio as our main effect size metric: $RR_x = \ln(E/C)$, where E is A_{max} in the enriched or high nutrient treatment, C is A_{max} in the un-enriched control or low nutrient treatment (Elser et al. 2007). We then tested statistically whether RR_x for each treatment (RR_N, RR_P, $RR_{Combined}$), was significantly different from zero. In comparisons across multiple experimental systems where species, soils and environmental conditions vary considerably, the analysis of change relative to control is generally considered to be more meaningful than absolute differences between means (Elser et al. 2007).

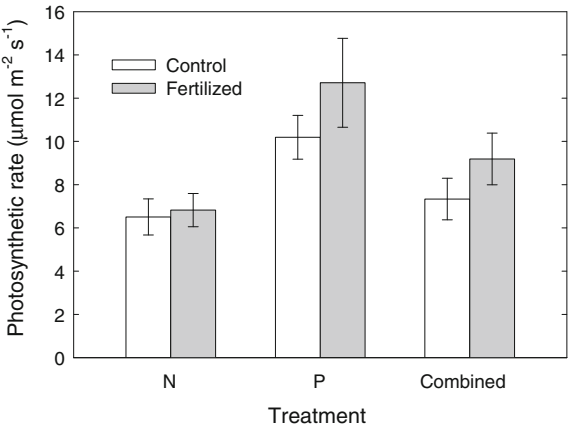


Fig. 2 Mean values of maximum photosynthetic rate under control versus nutrient addition treatments for tropical tree species fertilized with N, P, or combined nutrient treatments

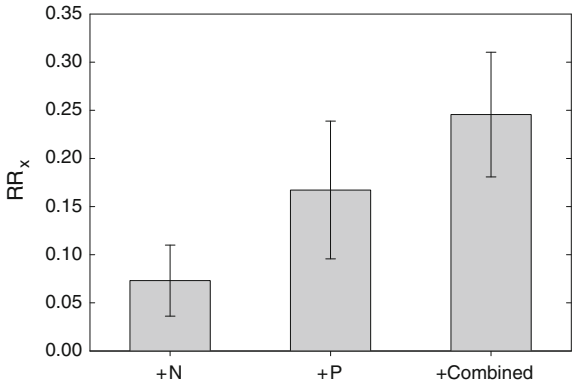


Fig. 3 Responses of area-based maximum photosynthetic rate to single enrichment of N, P or to combined elemental enrichment in tropical woody species. Data are natural-log transformed response ratios (RR_x) in which photosynthesis in the enriched treatment is divided by its value in the control treatment and then *ln*-transformed (Elser et al. 2007). For example, a value of 0.5 indicates a value in the manipulated treatment that is approximately 1.6 times its value in the control. *Sample sizes* +N (22), +P (21) and +Combined (24). *Error bars* indicate plus or minus one standard error

Our synthesis indicates that nutrient limitation of photosynthesis exists across the woody tropical species studied thus far (Fig. 3). However, the magnitude of the response varied strongly with the element in question. Across the 22 species × site combinations for N, photosynthesis trended higher with N addition, but mean RR_N was not significantly different from zero at the 0.05 significance level ($p = 0.061$ for *t*-test of $RR_N = 0$). Mean RR_P was significantly different from zero, indicating that photosynthesis significantly increased with P across the 21 species × site combinations ($p = 0.03$ for *t*-test of $RR_P = 0$). Thus our results across studies are consistent with the idea that P is the most widespread limiting nutrient in tropical forests. Whereas N produced substantial species-specific responses at some sites, with RR_N values up to 0.50 indicating an increase of approximately 1.6 times its value in the control, the magnitude of photosynthetic response to N was muted compared to P, which showed RR_P values as high as 0.90 indicating an increase of 2.45 times its value in the control treatment.

Our data also indicate that the 24 species × site combinations with combined manipulation of N, P, K or micronutrients produce the greatest response across all studies ($p = 0.0009$ for *t*-test of $RR_{Combined} = 0$; Fig. 3). $RR_{Combined}$ ranged as high as 0.92 indicating an increase of 2.51 times its value in the control. The highest significance in the case of combined nutrient addition leads to one of two possible conclusions. The first is that synergistic effects of multiple elemental manipulation has a stronger impact on alleviating nutrient limitation of photosynthesis than single element manipulation, consistent with the theory of multiple limiting nutrients (Kaspari et al. 2008). This theory suggests that plants allocate to uptake of limiting

resources until all resources simultaneously limit a physiological process such that addition of any resource increases the rate of that process to the same degree (Bloom et al. 1985). Alternatively, the greatest response to combined nutrient addition could result from the fact that manipulating multiple elements leads to a greater chance that the single most limiting element is included in the study. Overall, the high values for $RR_{Combined}$ support the idea that the stoichiometric demands of fundamental biochemical processes enable co-limitation by multiple elements (Elser et al. 2007; Kaspari et al. 2008; Wright et al. 2011; Santiago et al. 2012).

Although our mean RR_x values were significantly different from zero or nearly so in all cases, a closer look at individual species × site combinations reveals that RR_x values were negative in 21 out of 67 cases. Negative RR_x values were fairly distributed across nutrient treatments, with eight negative RR_N values, six negative RR_P values, and seven negative $RR_{Combined}$ values. Furthermore, 11 out of the 21 negative RR_x values were from a single study, in which 11 out of the 12 RR_x values in the study were negative (Saraceno 2006). The study by Saraceno (2006) is in many ways unique because it was conducted in Brazilian cerrado sensu stricto vegetation, an open canopy savannah ecosystem, whereas most of the other studies include tree species of closed canopy rainforest ecosystems. Cerrado regions are known for nutrient poor soils that are acidic with a low cation exchange capacity and high concentrations of aluminum (Furley and Ratter 1988). Additionally, light availability is likely to be higher in cerrado because of the open canopy and rainfall is more seasonal than in many closed canopy rainforest ecosystems. Whereas most rainforest species increase photosynthetic rate with the addition of nutrients, cerrado trees nearly uniformly decrease photosynthetic rate (Saraceno 2006), and reduce both nighttime and daytime transpiration with the addition of nutrients (Bucci et al. 2006; Scholz et al. 2007).

Several studies have directly addressed the responses of tropical trees to elevated atmospheric CO_2 by experimentally combining elevated CO_2 treatments with nutrient addition. Application of soil fertilizer substantially accelerated biomass accumulation of two tropical tree species at elevated CO_2, but elevated CO_2 had no significant effect on biomass accumulation in unfertilized communities (Winter et al. 2001), providing support for nutrient limitation of the concentration-carbon feedback. However, relatively little data is available on photosynthetic responses to nutrients at high CO_2 (Fig. 4). In one study from southeast China, N fertilization did not modify the effects of CO_2 on photosynthesis (Liu et al. 2012). Yet, in two other studies from the neotropics, greater combined nutrient availability increased the magnitude of the response of photosynthesis to elevated CO_2 (Carswell et al. 2000; de Oliveira et al. 2012), providing support for nutrient limitation of the concentration-carbon feedback. The results of these studies are consistent with the conclusion that N limitation of photosynthesis in tropical trees is modest compared to P, and combined nutrient treatments produce the greatest responses (Fig. 4).

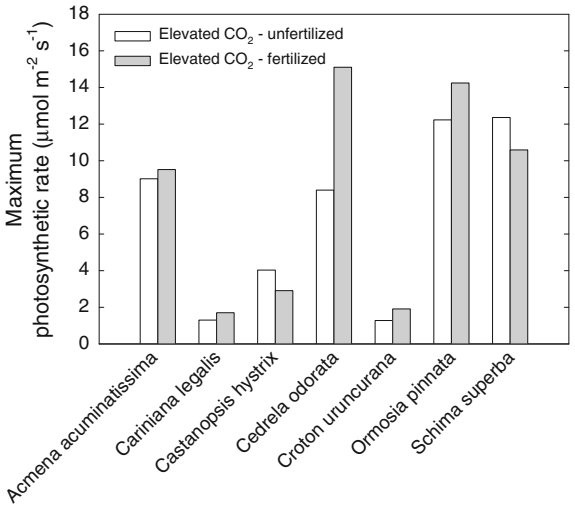


Fig. 4 Maximum photosynthetic rate at elevated CO_2 for tropical trees in control and fertilized treatments

Conclusions and Future Directions

We sought to determine whether nutrients limit the concentration-carbon feedback, the ability of tropical trees to increase rates of photosynthetic carbon gain as atmospheric CO_2 increases. Our evaluation of the available literature indicates several key patterns. First, nutrient limitation of photosynthesis is widespread in rainforest trees and appears to manifest itself differently, if at all, in savannah trees. Second, although photosynthesis is strongly limited by N in particular species, N limitation of photosynthesis is modest compared to P limitation of photosynthesis when considering all of the available literature. Finally, alleviation of nutrient limitation through addition of combined nutrient treatments produces the strongest photosynthetic responses, highlighting the potentially complex stroichiometric interactions among elements.

Currently, our predictions for how tropical forest ecosystems will respond to climate change and how their responses will feedback to the global climate system are limited by model realism. Several models still represent photosynthesis as continuously increasing in proportion to atmospheric CO_2 (Bonan and Levis 2010), illustrating the need for improved representation of photosynthesis in Earth system models (Ali et al. 2015). To build on the current state of knowledge regarding nutrient limitation of photosynthesis in tropical trees, an expansion of approaches will be necessary. First, the focus on N and P needs to be broadened to include other potentially limiting elements. Only one long-term tropical fertilization study has added K in a factorial design, but it reported substantial K limitation of key ecosystem processes (Wright et al. 2011). Also, given the diversity of tropical soils, one wonders if there are other elements beyond N, P, and K that deserve focused study. Second, better correspondence between large-scale field experiments and key

greenhouse experiments designed to test specific mechanisms are needed to pinpoint physiological responses and refine their role in processes at larger scales. Finally, a greater distinction between nutrient limitation of photosynthetic productivity between early and late successional forests is needed to elucidate the potential carbon sequestration of secondary forest during regeneration. Secondary forests are becoming a larger proportion of the total tropical land area, and their rapid growth during early succession has been implicated as a potential major route for carbon absorption by terrestrial vegetation (Cernusak et al. 2013). This potential must be determined to improve our predictive power. Overall, understanding the potential and limits to the key carbon cycling processes of tropical trees will improve our ability to forecast global responses and feedbacks to climate change.

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